Building reports and dashboards

Report and dashboard design principles

Overview of the design principles

Most of the **general design and storytelling principles and practices** on the use of colour, white space, proximity, purpose, etc. are directly applicable to dashboards and reports.

Define the story and objective

Begin by **understanding what the story** and **objective** are, who the **audience** is, and what their requirements are. We use this to **plan the layout** of our dashboard and **what we want to visualise**.

Clear and intuitive layout

Arrange visuals logically and aesthetically. Use a **clean layout**, placing important visuals prominently and arranging them in a **meaningful flow**.

Ensure interactivity

Incorporate **slicers**, **filters**, **buttons**, and **bookmarks** for interactivity, while ensuring they're intuitive and align with our report's objectives.

Overview of the design principles

Limit clutter and noise	**Avoid overcrowding** the report with **unnecessary visuals or information**. Less is often more; we need to focus on what's **most relevant**.
Visual hierarchy	Use **visual hierarchy to emphasise key metrics and insights**. Use font sizes, colours, and positioning to **guide the user's attention** to the most important information.
Consistent design and branding	**Maintain consistency** in design elements, such as colours, fonts, and logos, to ensure a cohesive and professional appearance.
Ensure accessibility	Ensure the data, visualisations, and reports are easily accessible and understandable. **Use clear titles**, **labels**, **descriptions**, **markers**, **alt text**, etc. to aid comprehension.

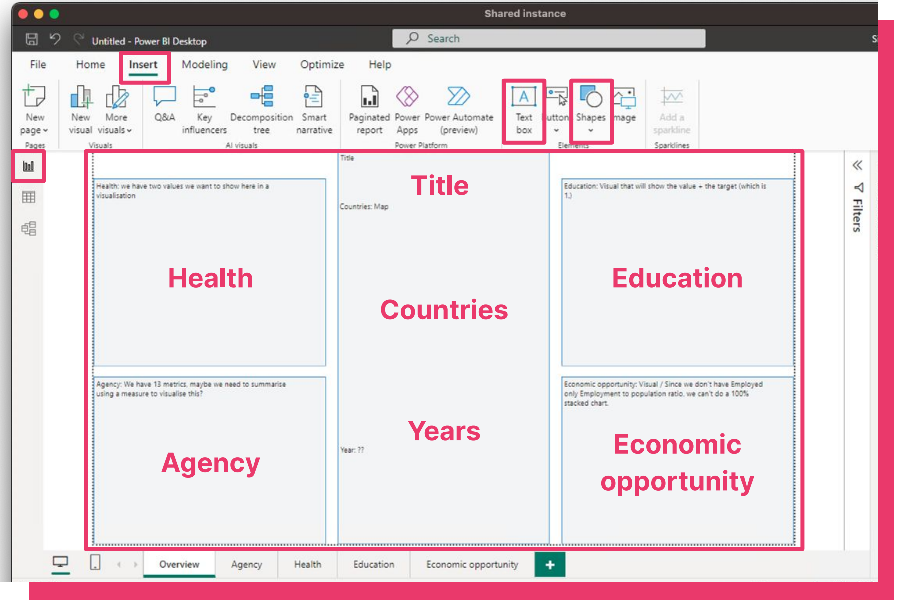
Understand what the story is we need to tell

Since we now know how to wrangle data and create visualisations in Power BI, it's important that we **understand what the story and objective are before building a dashboard or report**.

We can use any tool to create a draft layout of our report.

We can also **use shapes and text boxes** in our **Report view** to map out our layout.

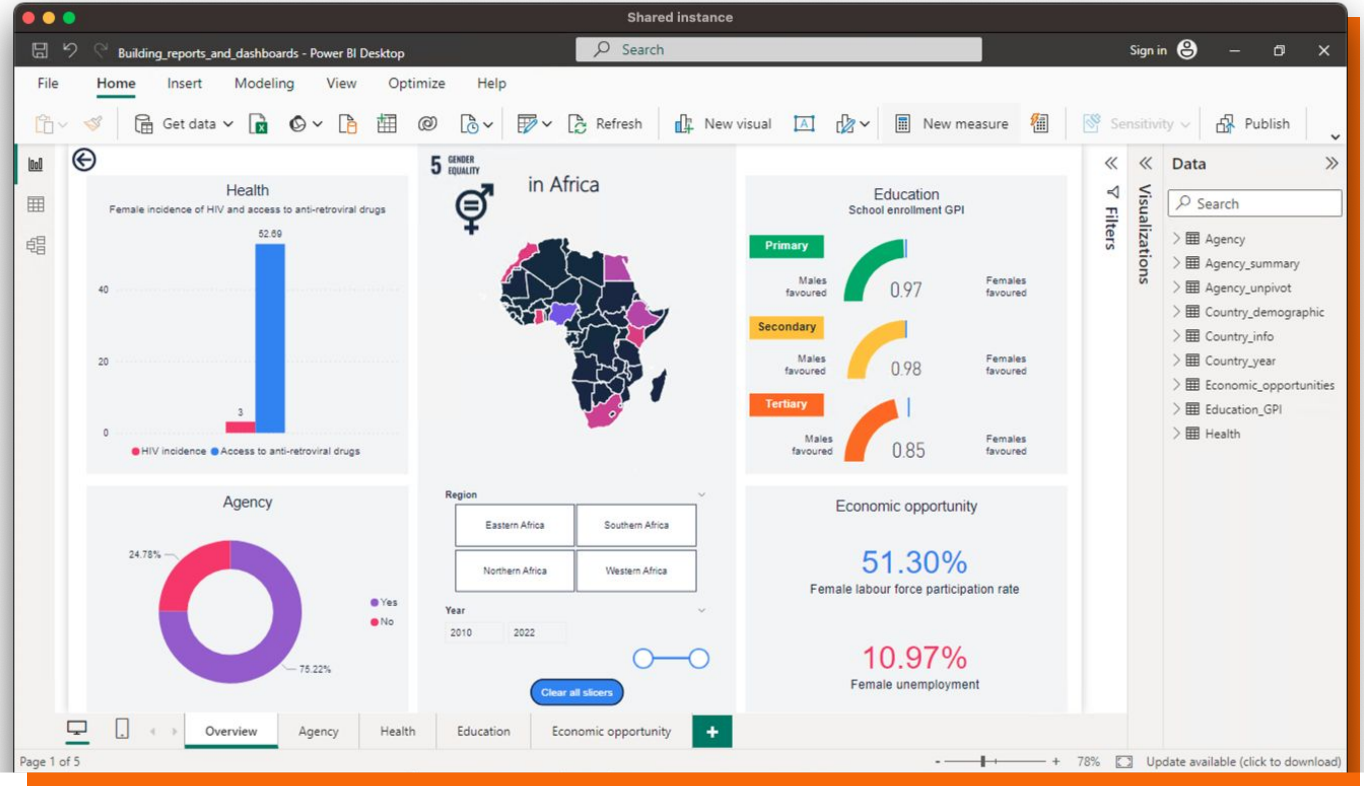
By using this approach, we can **plan** how we will **use the space available** on the page, ensure that the **elements do not overlap**, and that we have **included all the elements** we want to add.



A clear and intuitive layout

Because we've planned, it's much easier to see that our layout is clear and intuitive before we start creating any visualisations.

We've **grouped related elements** and added the most **prominent visuals and interactions together** to enable simple navigation.



Ensuring interactivity

We can **ensure interactivity** by including multiple features:

01. Filters

02. Slicers

03. Buttons

04. Tooltips

05. Drills

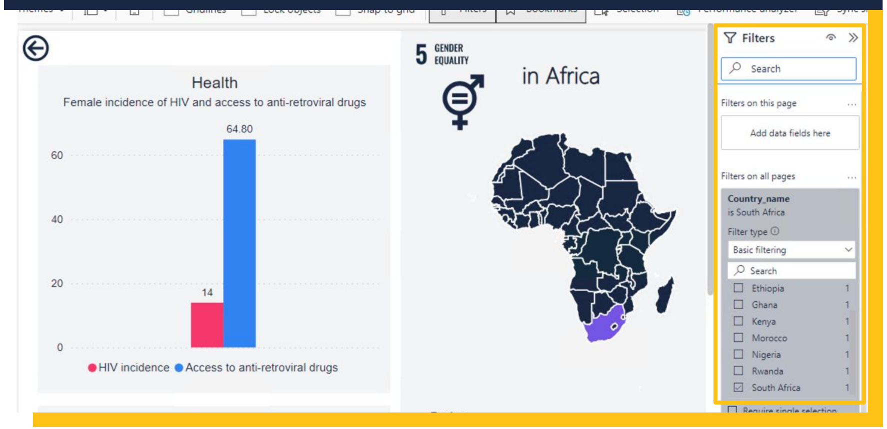
06. Bookmarks

Ensuring interactivity

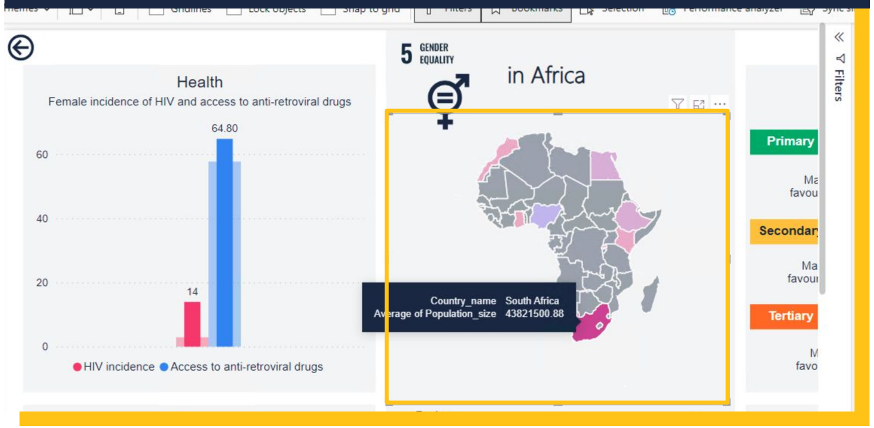
01. Filters

In Power BI, we can **filter** by using the **Filters pane**, a **visualisation**, or a slicer. Each of these methods serves a specific purpose but all three allow for dynamic filtering and/or highlighting.

The **Filters pane essentially "removes" data** from the report pages. This method is **useful to remove values that we don't need** in a specific visualisation, page, or report.



When we use an **element to filter with**, such as a visualisation, then the **filtered data are highlighted and nothing is removed**. It is useful when we want to **compare filtered values to a total**.



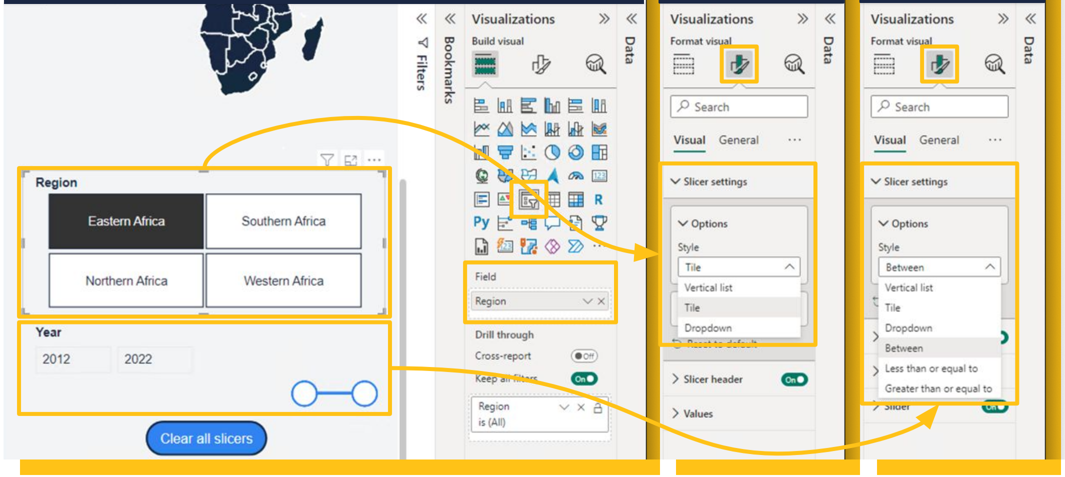
Ensuring interactivity

02. Slicers

A slicer is **a visual control element** that allows us to **interactively filter data** within a report. It presents a **list or table of selectable values**, usually from a specific column in the dataset, enabling us to **filter visuals based on our selections**.

The **type of slicer** we can use will **depend on the type of data** we have.

Categorical data | **Continuous** data



Slicers are useful:

- To give users the choice to **change the focus of the visuals**.
- **For simplifying** the ability to **see the current filtered state** without having to open the Filters pane.
- **For filtering by columns** that are not visualised.
- For creating a **more focused** report.

Ensuring interactivity

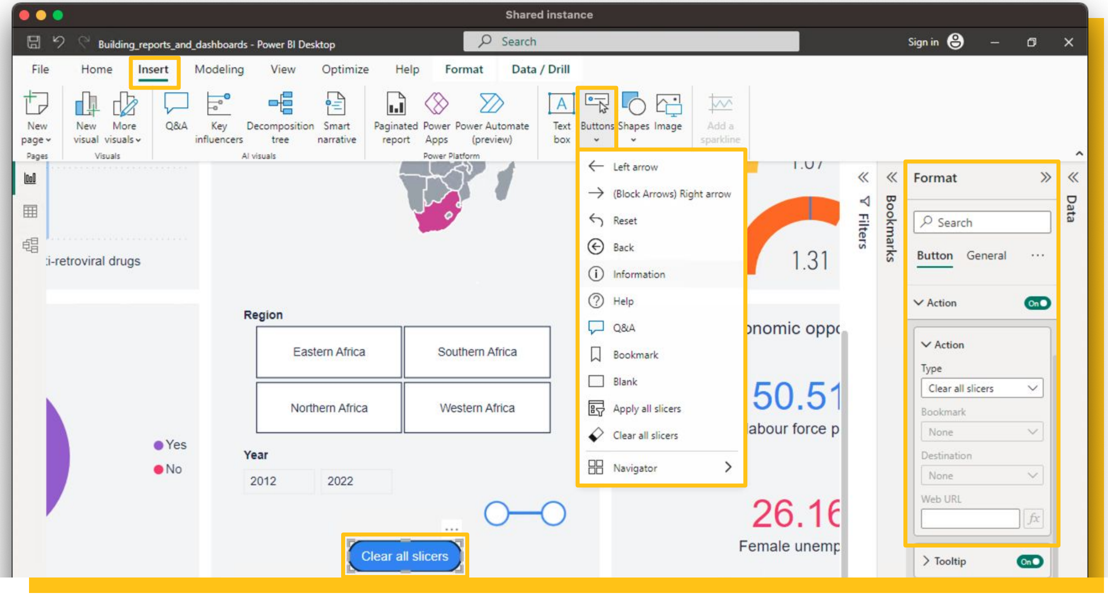
03. Buttons

Buttons are **interactive elements used to trigger actions** within a report, enhancing the user experience and interactivity. They allow us to **perform predefined actions or navigate** through the report based on their selections or clicks.

Customisable actions: Can be configured to perform various actions, such as navigating to different report pages, clearing filters, and other actions.

Visual elements: Can be customised with different shapes, text, colours, and icons to make them align with the page styling.

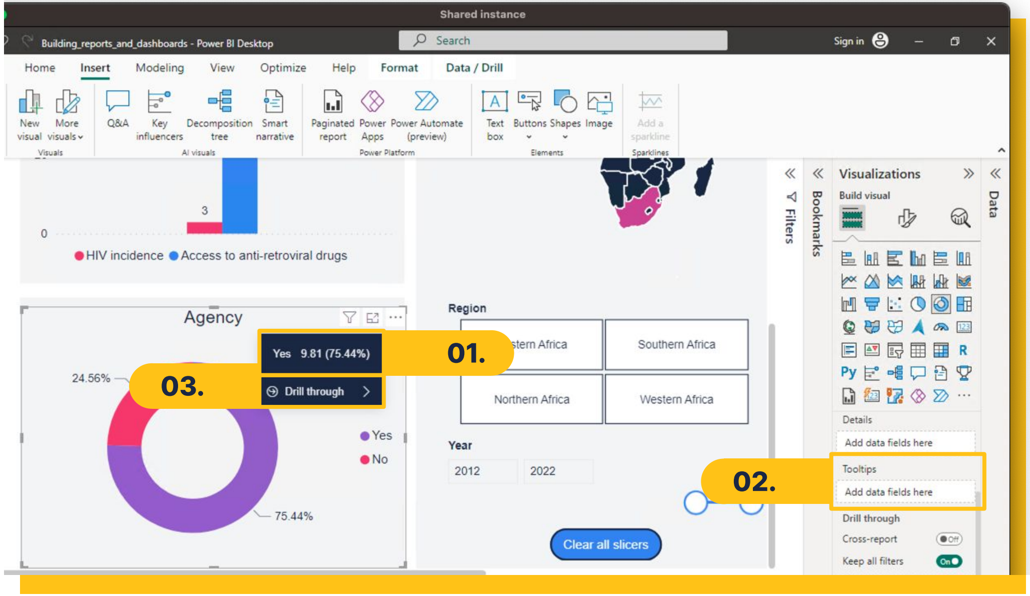
Multiple interactions: Can be set to respond in a variety of ways, such as hover effects and click actions.



Ensuring interactivity

04. Tooltips

Tooltips **provide additional context and details** for visuals in a report when we **hover over or interact with specific data points or elements**. These additional details offer **more in-depth information** without cluttering the main visual.



01. Tooltips will **automatically display** when we hover over a visualisation. The **details** within the tooltip will **depend on the feature(s)** represented in the visualisation.

02. We can **add additional information** by adding more features to the **Tooltips well**.

03. We can **navigate through drills** (through, up, and down) through the tooltip, but only if it is **activated in the settings**.

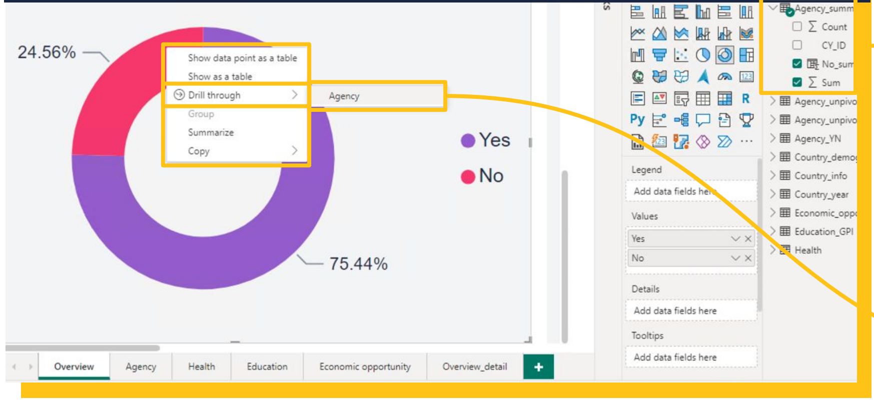
Ensuring interactivity

05. Drills

In Power BI, drills encompass various functionalities that allow us to **navigate through different levels of data or details** within a report. These include drill-through, drill-up, and drill-down capabilities.

Drill-through: Allows us to **access more detailed information** about a specific set of data points by drilling through it to a **different report page**.

We can **access** the drill-through option through the **modern tooltip** or by **right-clicking** on a visualisation.

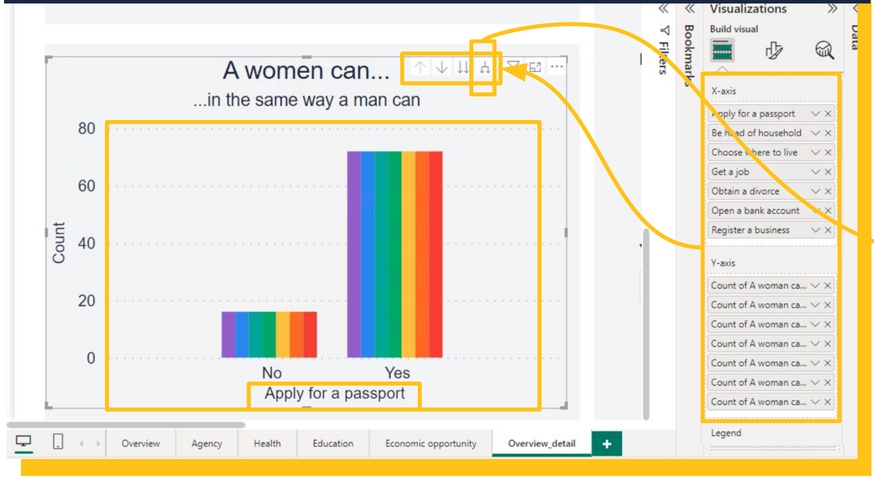


To **enable** drill-through, we need to **create the related report page** to which the drill will allow us to navigate, and **add the columns to the drill-through well** to this page.

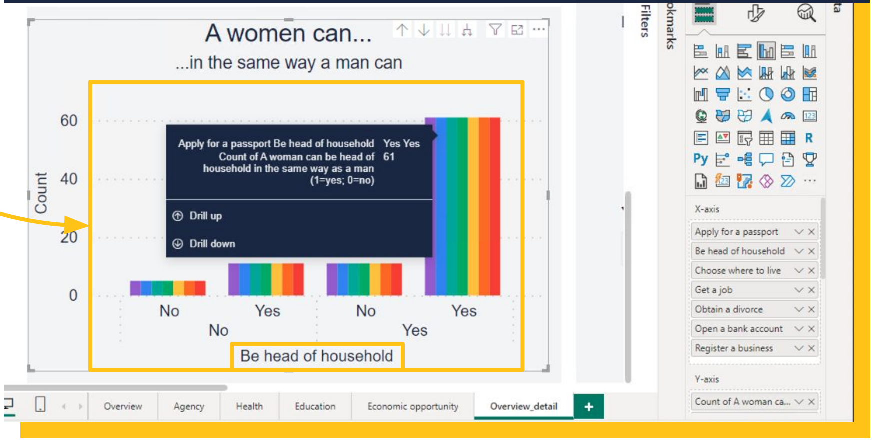
Ensuring interactivity

Drill-up and drill-down: Allows us to **navigate through the different hierarchical levels** of a dataset within the same visualisation and report page. When we **drill up**, we **navigate to a higher level** of the data. When we **drill down**, we **navigate to a more granular level** by breaking down summarised data.

Drill-up and down are **only available** when we have our data set up in a **hierarchical** way. The drill **icons** will then **appear at the top right** of a visualisation when we hover over it.



We can drill-down **into a specific data point** or **one level down** in the hierarchy, or simply **move to the next level**. Our **visualisation will change** and we can **continue to drill-down**, depending on the number of features in the visualisation.

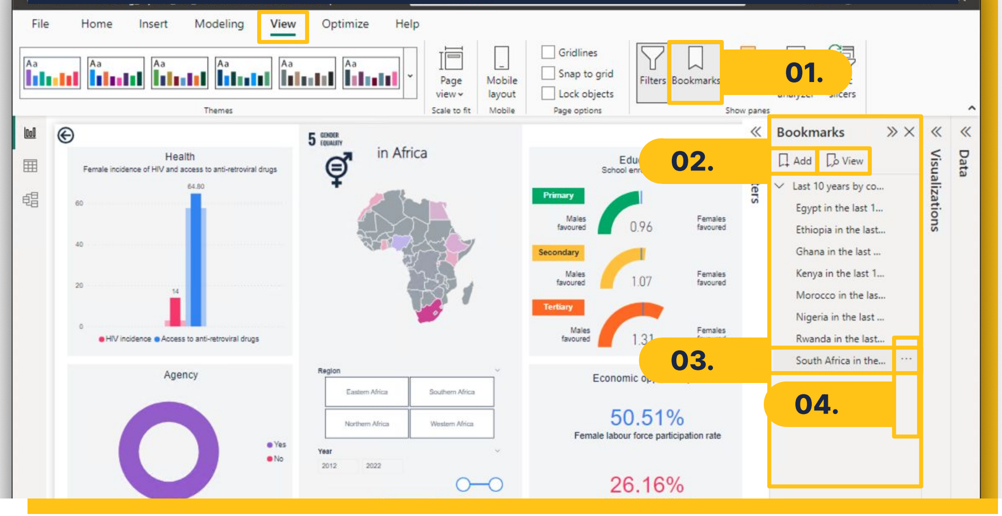


Ensuring interactivity

06. Bookmarks

Bookmarks enable us to **capture and save specific views** or states within a report, allowing for interactive navigation and storytelling.

Example bookmark: “South Africa in the last 10 years”
Report sliced for the country South Africa and the years 2012 to 2022.



01. To **enable** the Bookmarks feature, go to View and select Bookmarks.

02. In the Bookmarks pane on the right, we can **add and view our saved views**.

03. Views are a saved version of the report with **some filters and slicers already applied**.

04. We can **change** the slicers and filters on the report and **click on the ellipse to update the bookmarked view**.

Key points to remember

Planning and an **intuitive and simple layout** play a significant role in the success of a report.

While the **interactivity features can be useful**, they need to be used **intentionally**, otherwise they can distract more than enhance.

Consistent and **visually pleasing designs and visualisations** can significantly contribute to the professionalism of a report.